REFERENCE CARD · REV. AG

A5K — Technical Specifications

Clocks, memory, processors, bandwidth and the limits they impose

Rev. AG · 2 November 2026 · Companion to the design proposition · Figures are targets, not measurements

> **Design study.** Nothing described here has been built or measured. Clock rates are synthesis estimates for the named device families; bandwidth figures are arithmetic from bus width and data rate; processor core counts are what the logic can hold, not what the software needs.

1. What this is — and what it is not

This document specifies A5K (C+), the frozen reference model. It is a PCIe add-in card. It is not a computer, not a single-board machine, and not a replacement for a motherboard. It has no CPU, no boot firmware and no operating system of its own. It is a display and timing device that plugs into a host PC, in the same mechanical and electrical class as a graphics card.

Table S1. Product classes, and which one exists.

Product	Status	Requires	Provides
A5K card	**the current design**	An x86-64 host with a free PCIe slot	Chipset, Chip RAM, HDMI scanout
A5K module (SoM)	not designed	A carrier board	The same chipset in a socket
A5K mainboard	mentioned as a form factor only	Mains power and a case	A complete machine

The host supplies the CPU, the memory the OS runs in, the storage and the operating system. The card supplies the chipset, its own Chip RAM and its own display output. Unplug the host and the card is a board with a power LED.

The small RISC-V core on the card is not an application processor. It is housekeeping — timers, GPIO, the SD card and interrupt aggregation — running bare-metal firmware at 50–150 MHz, with no MMU and no loader. Applications and drivers run on the host, compiled for the host's architecture. The card contributes the chipset, not the CPU.

It cannot be plugged into a Commodore Amiga. The A2000 has Zorro II, the A3000 and A4000 have Zorro III, the A1200 has a trapdoor and PCMCIA. None of them has PCIe, and no bridge exists that would make a modern card enumerate on a 68k bus.

2. Host requirements

Today the card requires an **x86-64 PC**. Other host architectures are technically possible — the interface is standard PCIe — but they are untested, and some RISC-V boards expose only a single lane.

Table S2. Minimum host, and the recommended build.

Item	Minimum	Recommended
Architecture	x86-64	any recent desktop CPU

Item	Minimum	Recommended
Processor	2 cores at 2 GHz	4 cores, anything from the last decade
System RAM	4 GB	8 GB or more
Free slot	PCIe ×1 electrical or larger, Gen2+	×4 or ×16 length, Gen3
Slot power	25 W from the slot, no auxiliary connector	as minimum
Firmware	UEFI with Above 4G Decoding enabled	Resizable BAR also enabled
Operating system	Linux 6.x with DRM/KMS	a current distribution
Disk	20 GB free for tooling	—
Displays	one monitor for the host, one for the card	two inputs on one monitor
Case	clearance for a low-profile card	full-height bracket

> **Notes.** A laptop requires a Thunderbolt-to-PCIe enclosure, because the card needs a real slot. Above 4G Decoding must be enabled in UEFI for the 64-bit BAR that maps Chip RAM; on some consumer boards this is the only setting that has to be changed. The card draws roughly 5 W, so a ×4 slot is ample and no power cable is needed.

3. Why ×1 on the reference card, and ×4 on the Studio Pro

The reference card is **electrically ×1**. The display is fetched from the card's own DDR3L; PCIe carries Copper lists, register writes and the occasional frame readback. Half a gigabyte per second is already an order of magnitude more than control traffic needs. The Studio Pro uses ×4 because two pipelines and a capture stream want real bandwidth.

Table S3. Slot options, and what each would cost.

Slot	Lanes	Gen2 / Gen3	Fits	Cost
×1	1	0.5 / 1.0 GB/s	every slot, including ×1	**C+ reference**
×2	2	1.0 / 2.0 GB/s	×2 and larger slots	Studio variant
×4	4	2.0 / 3.9 GB/s	×4, ×8 and ×16 slots	**Studio Pro**
×8 / ×16	8 / 16	4–16 GB/s	rare on desktops	not used

The lane budget decides it. An Artix-7 100T has exactly four GTP transceivers, and 4K60 output consumes three of them for HDMI. One is left, and that is the PCIe link: ×1. The Studio Pro moves to a Kintex-7 325T, where sixteen transceivers allow two 4K60 transmitters and ×4 to the host.

A ×1 card still fits any slot. A ×1 card seats in ×1, ×2, ×4, ×8 and ×16 connectors, including the ×4 slot a desktop usually has free. The physical connector on the reference card is ×4-length for mechanical stability; the link is ×1.

4. Clock domains

Table S4. The domains a card would run in.

Domain	Rate	Purpose
Copper / chip bus	7.0714 MHz (141.4 ns)	One instruction per tick; exactly 21 slots of the 148.5 MHz bus. The machine's timing contract
Pixel clock, 1080p60	148.5 MHz	Scanout

Domain	Rate	Purpose
Pixel clock, 4K60	594 MHz (4 lanes)	Scanout, above the fabric's comfort zone
DDR3L	800 MHz (1600 MT/s)	Chip RAM
PCIe reference	100 MHz	Link training and packet clock
Host CPU	external	Whatever the host provides

5. Memory

Table S5. Chip RAM configurations.

Configuration	Devices	Capacity	Bandwidth	Note
Base	2 × 4 Gb DDR3L, 32-bit	1 GB	6.4 GB/s	as costed in the BOM
Upgrade	2 × 8 Gb DDR3L, 32-bit	2 GB	6.4 GB/s	same package, drop-in
Maximum	4 × 8 Gb, two ranks	4 GB	6.4 GB/s	top of the 32-bit space
Architectural ceiling	32-bit byte addressing	4 GiB	—	~3.5 GB usable after registers and apertures
64-bit variant	4 × 16 Gb DDR4	8–32 GB	12.8–25 GB/s	wider bus and 40-bit pointers; new part

Why DDR3L, and not DDR5. The target families cannot drive DDR5 at all — it needs an UltraScale+ or Versal part, at roughly ten times the FPGA cost. DDR3L is still in production for industrial and networking equipment, is stocked by every distributor, and costs about $7 per 512 MB. A 32-bit DDR3L bus supplies 6.4 GB/s, which is more than the slot table can consume. DDR5 would solve a bandwidth problem this machine does not have, at a price it cannot pay.

Upgrade path. 1 GB to 2 GB is a device swap in the same footprint — 4 Gb parts for 8 Gb. 2 GB to 4 GB needs two ranks and rank support in the memory controller. Beyond 4 GiB the obstacle is not the memory but the pointer width: the 64-bit variant widens both, and moves to DDR4.

> **ECC** is the more valuable upgrade for instrumentation work than capacity: on a 32-bit bus it means 40 bits and five x8 devices instead of two x16. DDR3 tops out at 8 Gb per device, which is why 4 GB is the wall for the 32-bit design. Host memory is reached by DMA over PCIe and is not limited by the card.

6. Processors

There are two CPU positions in this design, and they are not interchangeable.

Table S6. Processor options, clock and cores.

Option	Kind	Max clock	Cores	Where it runs
VexRiscv on ECP5-85	soft	~50 MHz	1–4	On the card; housekeeping
VexRiscv on Artix-7 200T	soft	100–150 MHz	1–8	On the card; practical sweet spot
Microblaze on Artix-7	soft	~200 MHz	1–8	Vendor core, closed toolchain
Rocket / CVA6	soft	50–100 MHz	1–4	Needs a larger part
PolarFire SoC	hard	667 MHz	4 + 1	Hardened cluster beside the fabric
x86-64 host	hard	4–6 GHz	4–256	PC, outside the card
RISC-V host (P670 class)	hard	~2–2.5 GHz	1–16	Host board

> **The on-card core is a supervisor, not an engine.** It runs CIA timers, GPIO and interrupt aggregation. Whether it runs at 50 MHz or 667 MHz changes almost nothing, because it does not draw

> pixels — the Copper, blitter and planes do. Spending fabric on clock speed here is spending it in the wrong place. The heavy work belongs on the host, on the other side of PCIe.

7. Bandwidth budget at 4K60

3840 × 2160 = 8.29 megapixels. The 32-bit DDR3L bus supplies 6.4 GB/s of raw memory, but the **64-bit slot fabric supplies only 1.19 GB/s** — the fabric, not the memory, is the limit, and it is the same on every model. The slot table must also fit the Copper, the blitter and audio into the same schedule. Slot width differs by model: **64 bits on C+** (1.19 GB/s) and **256 bits on the Studio models** (4.75 GB/s), which is what makes 10-bit HDR and 10-bit capture possible together.

Table S7. What 4K60 costs, per content type.

Content	Per frame	At 60 Hz	Of 1.19 GB/s	Verdict
8 planar bitplanes (1 bit/px)	8.3 MB	0.5 GB/s	42%	Comfortable
1 full-screen ARGB8888 plane	33 MB	2.0 GB/s	168%	Does not fit
2 ARGB planes	66 MB	4.0 GB/s	336%	No
4 ARGB planes	133 MB	8.0 GB/s	672%	No
8 ARGB planes	265 MB	15.9 GB/s	1,336%	No

> This is the quiet argument for planar graphics: at 4K60, eight bitplanes cost 42% of the bus and one full-screen ARGB plane costs 168%. The Amiga's bitmap format was a bandwidth strategy, not a limitation — and on this machine the limit is the slot fabric, not the DRAM behind it.

8. Parallelism

Table S8. What runs simultaneously, and what does not.

Unit	Count	Nature
Copper	1	One sequencer, one instruction per tick; parallel with everything else, serial within itself
Blitter	1	One operation at a time, by design
Planes	8 (16 optional)	Fetched in parallel each line
Audio channels	4–8	Independent DMA streams
Slot table	1	Fixed schedule; every cycle allocated before the frame
Host cores	unbounded	Outside the card, over PCIe

Parallelism here is spatial, not threaded. Roughly twenty independent units run at once because each does one thing forever at a fixed rate — not because anything is scheduled. Adding CPU cores does not add throughput; it adds contention on the same memory. The bound is bandwidth: at 1.19 GB/s of slot fabric, the display pipeline is the budget, and the slot table is the ledger.

9. Links and ceilings

Table S9. The buses that bound the design.

Link	Bandwidth	Binds
PCIe Gen2 ×1	0.5 GB/s	Host ↔ card, A5K (C+)
PCIe Gen2 ×4	2.0 GB/s	Host ↔ card, Studio Pro
PCIe Gen3 ×4	3.9 GB/s	Host ↔ card, preferred

Link	Bandwidth	Binds
PCIe Gen4 ×4	7.9 GB/s	Host ↔ card, beyond current parts
DDR3L 32-bit	6.4 GB/s	Raw; the 64-bit fabric draws 1.19 GB/s of it
Address space	4 GiB	Chip RAM ceiling

What would have to change for more. More planes at 4K means a wider memory bus before it means more logic. A faster on-card CPU means a hard core, not a bigger soft one. More host bandwidth means Gen3 or Gen4, which means an Artix-7 with GTP or a larger part. Each is a different board, not a different setting.

10. Video modes, and the way back

Timings are generated from registers, so a mode is a row in a table, not a circuit. Three tiers matter:

Table S10. The floor, the default and the normal case.

Tier	Mode	Why
Floor	**640×480 @ 60**	The lowest signal HDMI and DisplayPort can carry — 25.175 MHz. It is also the modern equivalent of Amiga “hi-res”: 640 pixels wide. A 15 kHz Amiga mode cannot be put on either port at all.
Default	**1024×768 @ 60**	XGA: accepted by essentially every display since 1990, and what the card falls back to when there is no EDID to ask. The user may change it; the setting lives in the configuration flash.
Normal	display preferred, up to 4K60	On a 4K panel the EDID asks for 4K60, and that is what the card drives.

The ceiling is a pixel rate, not a list. Any CEA or VESA timing at or below 594 MHz that the EDID declares is legal: the standard video rates 23.976–60 on two slot clocks (148.35 MHz for the 1000/1001 family), high refresh to 1080p240 and 1440p144, and 4K at 60. **1080p240 costs the same as 4K60** — 497.7 megapixels per second either way — so the budget is per pixel rate, not per resolution.

No variable refresh. FreeSync, G-Sync Compatible and Intel’s Adaptive Sync only operate when the source offers them; a source that does not leaves the panel at fixed refresh, which every panel accepts. It is a ceiling, like no 8K.

Four rules keep it out of trouble. Never emit a timing the EDID does not declare; use the preferred timing at boot; implement DPMS D0/D3 only, waking on hotplug and never idling the link; and debounce hotplug for 100 ms, because monitors chatter on power-save. Those four are what separate a card that works with every panel from a card that works with the one on the bench.

First power-on. Read the EDID and use its preferred timing. No EDID, or nothing readable: come up at **1024×768 @ 60**. Every mode comes from the same registers, so adding one costs a row, not a redesign.

Recovery, for a blank screen. Four routes, layered, and all four ship:

1. **Host register.** PCIe configuration space is independent of video, so a driver can force defaults even when the display is wrong.
2. `SAFE_MODE` **on the microSD.** Drop the file on the card and power-cycle — it comes up at 1024×768 with no host involved.

3. **Button or jumper.** One second at power-on selects safe video; ten seconds restores factory defaults. About $0.15 for a bracket button, $0.02 for a jumper.
4. **Automatic.** No EDID, or no hotplug for ten seconds, and the card cycles 1024×768 → 800×600 → 640×480 until one locks — what a PC BIOS does.

Precedence at power-on: button or jumper, then the SD file, then the stored default, then the display's EDID, then the automatic cycle. The button does not violate the "no physical switch" rule — that rule is about the Auto/HDMI/DP selector, not about recovery.

11. Quick reference

Table S12. The numbers worth memorising.

Item	Figure
Product class	PCIe add-in card — requires a host PC
Host, minimum	x86-64, 4 GB RAM, free PCIe ×4 slot
Card power	~5 W from the slot; no auxiliary connector
Chip RAM, base	1 GB DDR3L, 32-bit, 1600 MT/s
Chip RAM, upgrade	2 GB — 8 Gb devices, same footprint
Chip RAM, maximum	4 GB, two ranks
Architectural ceiling	4 GiB, about 3.5 GB usable
Memory bandwidth	6.4 GB/s DRAM; 1.19 GB/s usable through the fabric
Copper tick	7.0714 MHz (148.5 ÷ 21), 141.4 ns per instruction
Scanout	4K60 at a 594 MHz pixel clock, four lanes
Host link	PCIe Gen2 ×1 (0.5 GB/s) on C+; ×4 on the Studio Pro
On-card CPU	VexRiscv, 50 MHz (ECP5) to 150 MHz (Artix-7)
Host CPU	anything with PCIe; 2–6 GHz, 4–256 cores
Planes	8, optionally 16
Outputs	4K60 SDR HDMI (primary), 1080p60 DP (secondary); HDMI sensed first. HDR10 on the Studio models
Mode floor	640×480 @ 60 — the lowest either port can carry
Refresh	Fixed only. 24–60 standard rates, 1080p to 240 Hz, 1440p to 144; no VRR
First power-on	EDID preferred timing; 1024×768 @ 60 when there is no EDID
Fabric	64-bit slots (C+), 256-bit (Studio), 512-bit reserved (Pro II) — all at 148.5 MHz
Blitters	1
Audio	8 channels, 24-bit; factory default 2.0 on every output; 4 jacks, TOSLINK, HDMI/DP LPCM; front-panel HP/mic header with its own DAC; the chosen mode is stored
4K60 cost, planar	0.5 GB/s for eight bitplanes
4K60 cost, ARGB	2.0 GB/s per full-screen plane

> Sources: datasheet figures for the ECP5-85 and Artix-7 200T families, DDR3L at 1600 MT/s, and arithmetic from bus width and pixel format. No figure here has been measured on hardware, because there is no hardware.